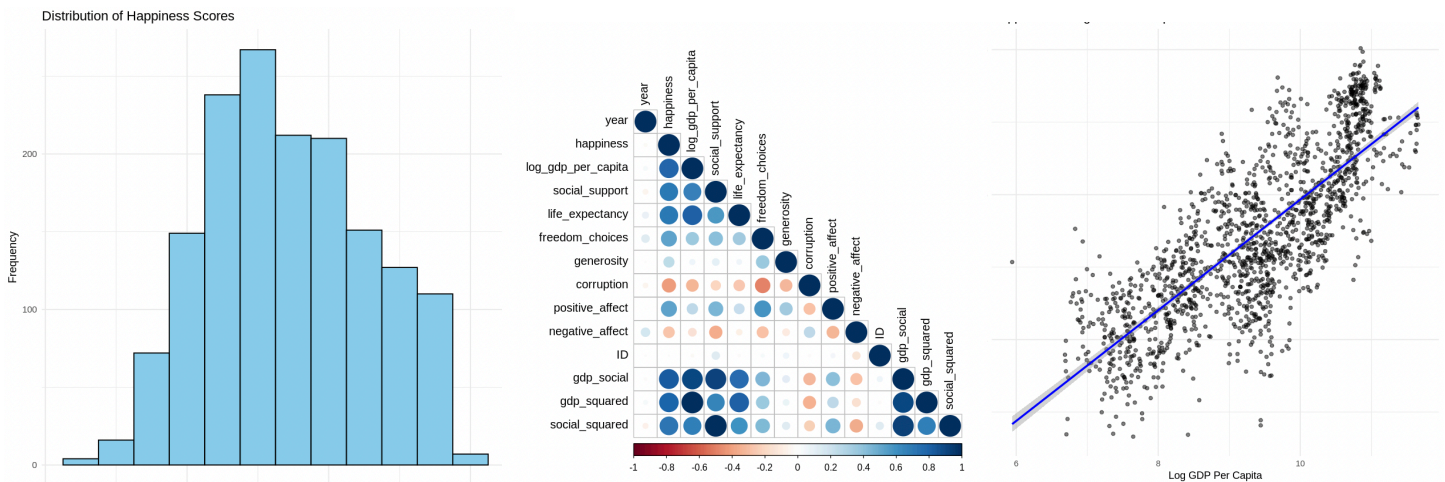


Exploratory data analysis (EDA)

The dataset includes various indicators such as log_gdp_per_capita, social_support, freedom_choices, and corruption. While some variables, like GDP and social support, exhibited clear linear relationships with happiness, others displayed more non-linear patterns. These observations made me explore a range of models to capture both simple and complex relationships effectively.

The histogram shows that happiness scores are normally distributed, centered around 5, with a slight right skew. Most countries fall within the mid-range of the happiness spectrum. The scatter plot shows a positive linear relationship between log_gdp_per_capita and happiness. Wealthier countries tend to report higher happiness levels. The correlation heatmap reveals a strong positive correlations between log_gdp_per_capita, social_support, and happiness, Other factors like life_expectancy and freedom_choices also show positive associations, Corruption has a negative correlation with happiness, as expected.



Building the Baseline Model: Linear Regression

I began with linear regression to model the relationship between happiness and the predictors.

$$\text{Happiness} = \beta_0 + \beta_1(\log(\text{GDP per capita})) + \beta_2(\text{Social Support}) + \dots + \beta_k X_k + \epsilon$$

the residual analysis showed that linear regression struggled with under-fitting, particularly for extreme cases such as countries with low GDP or high corruption. Additionally, multi-collinearity among predictors is there so I decided to use Elastic Net regression.

Elastic net regression combines Lasso (L1) and Ridge (L2) penalties to address multicollinearity and feature selection:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \left(\alpha \sum_{j=1}^p |\beta_j| + (1 - \alpha) \sum_{j=1}^p \beta_j^2 \right)$$

where α controls the mix between Ridge (L2) and Lasso (L1).

This model improved over linear regression, capturing key trends while addressing multicollinearity. Residuals shows that the model captures overall trend but struggles slightly with extreme values.

Non-Linear Relationships with XGBoost

As I went ahead with XGBoost, which is a tree-based model optimizing known for its ability to handle non-linear relationships and interactions.

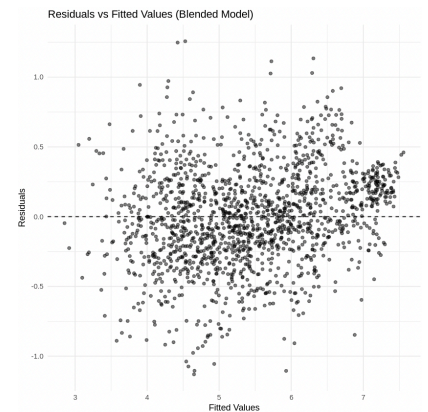
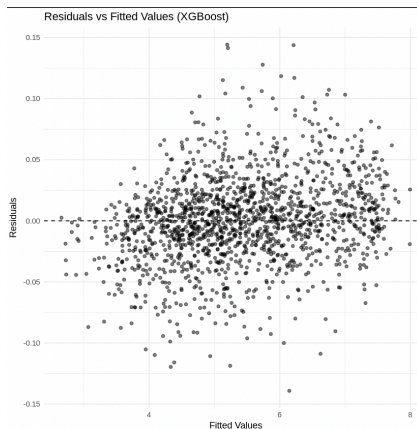
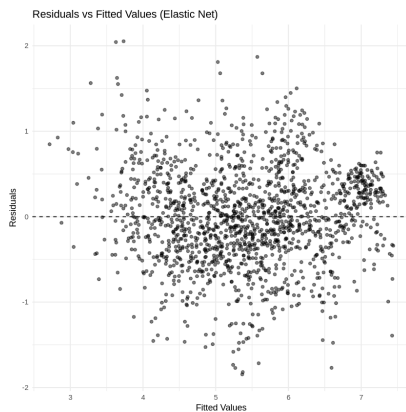
$$\text{Objective} = \sum_{i=1}^N \ell(y_i, \hat{y}_i) + \sum_{k=1}^T \Omega(f_k)$$

where ℓ is the loss function (squared error), and $\Omega(f_k)$ is the regularization term.

Key predictors (social_support, freedom_choices) remained dominant, but negative_affect gained importance. XGBoost outperformed Elastic Net due to its ability to automatically model interactions and handle non-linearities. Residuals are closer to zero compared to Elastic Net, indicating better predictions.

Blending Models for Final Predictions

So I decided to blend all the models to combine different things different models give the best of all and make sure that all aspects of data are captured. The final blended model combines Elastic Net and XGBoost. Blended residuals showed improved performance compared to individual models. The model effectively balanced linear and non-linear trends, reducing overall errors.



7.

Conclusion and Future Work

Elastic Net struggles with non-linearity but it is balanced by XGBoost which captures non-linear interactions. Blending models leverages the strengths of both, improving overall performance which can be observed in the residual plot.

Exploring advanced models like neural networks for non-linear relationships or Bayesian methods for uncertainty estimation might give better accuracy.